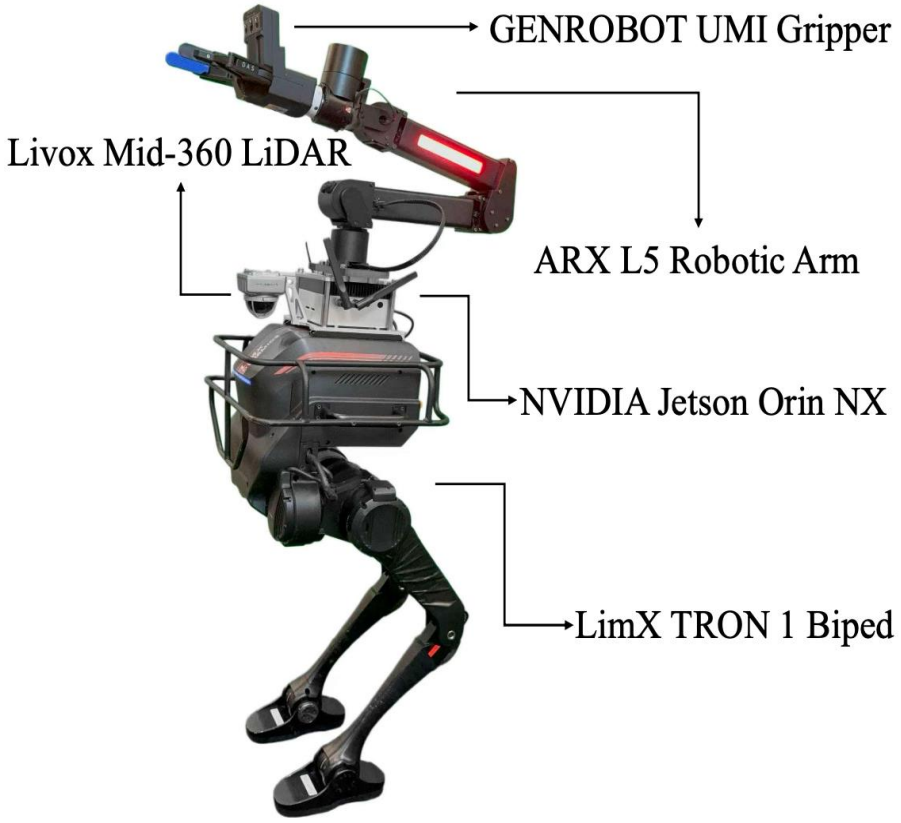
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Learning Holistic Whole-Body Loco-Manipulation with a Bipedal Mobile Manipulator

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Xudong Li, Zhizhan Li, Leixin Chang†, Hua Chen

Motivation

- Targets beyond the arm's nominal workspace require coordinated arm motion, postural adaptation, and stepping while maintaining balance.
- Existing interfaces often require explicit base-motion, gait, or footstep commands.



Goal: Whole-body control from a single 6-DoF end-effector target.

Key Contribution

Unified interface

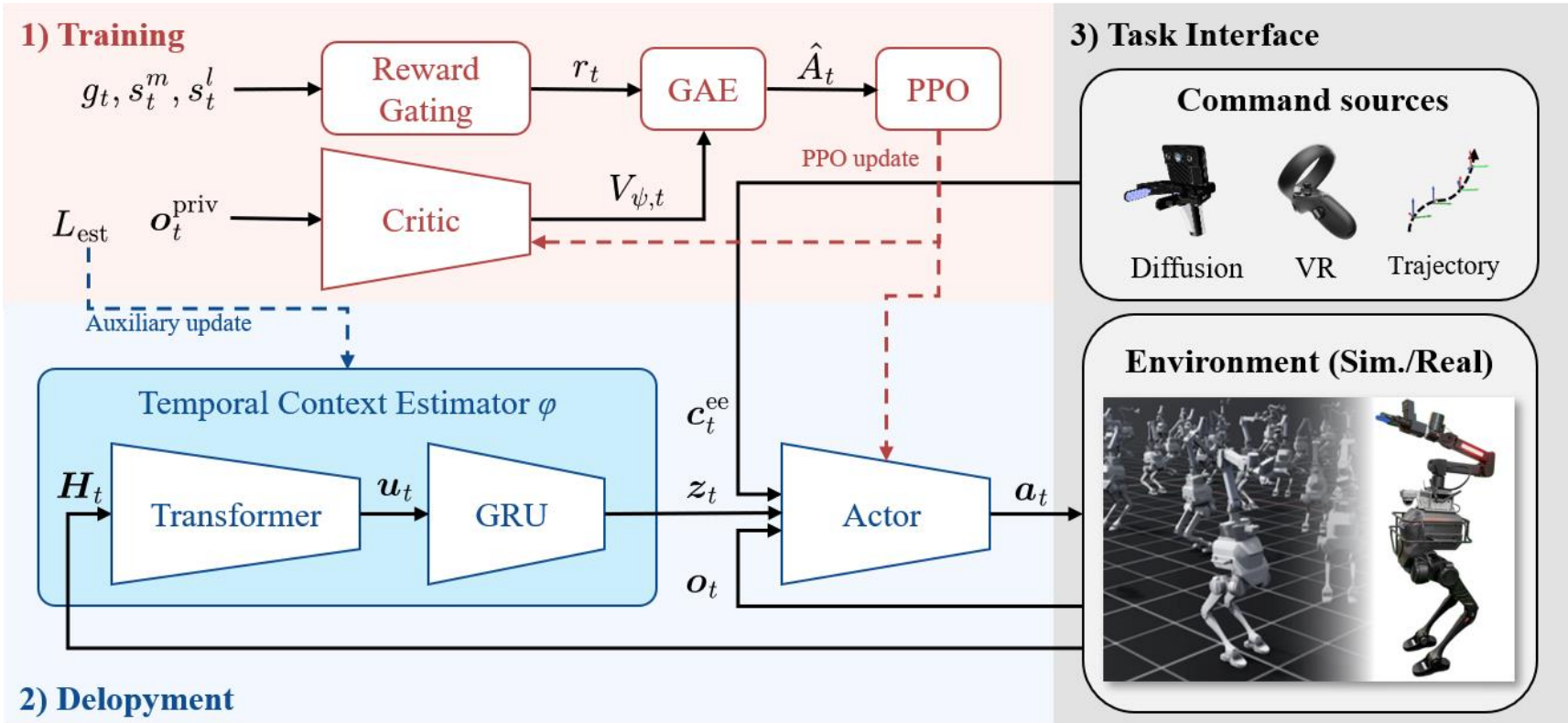
Maps a single 6-DoF end-effector target to commands for all 14 joints.

Reward Gating

Balances target approach with precise tracking and whole-body stability across different task phases.

Temporal Context Estimator

Combines a Transformer and GRU to infer hidden dynamics from observation history.



Main Results

- The full controller achieves an **88.30%** success rate in simulation.

- On the real robot, our controller extends the vertical reachable range to **3–191 cm** compared with 38–163 cm (floating-base + IK baseline).

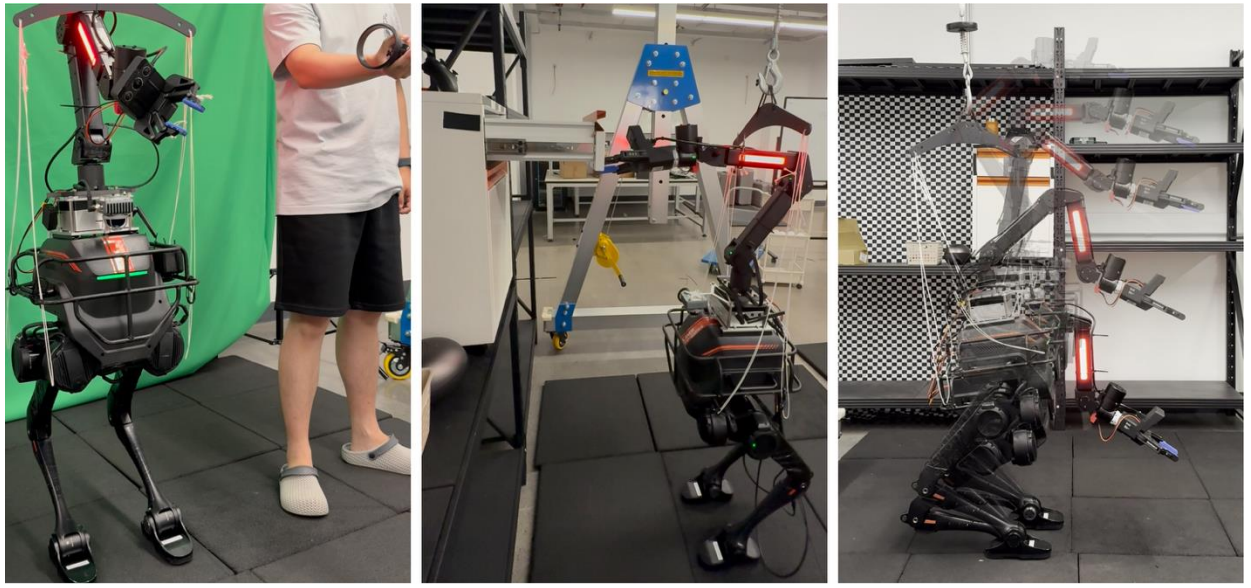


Fig1. Unified Interface

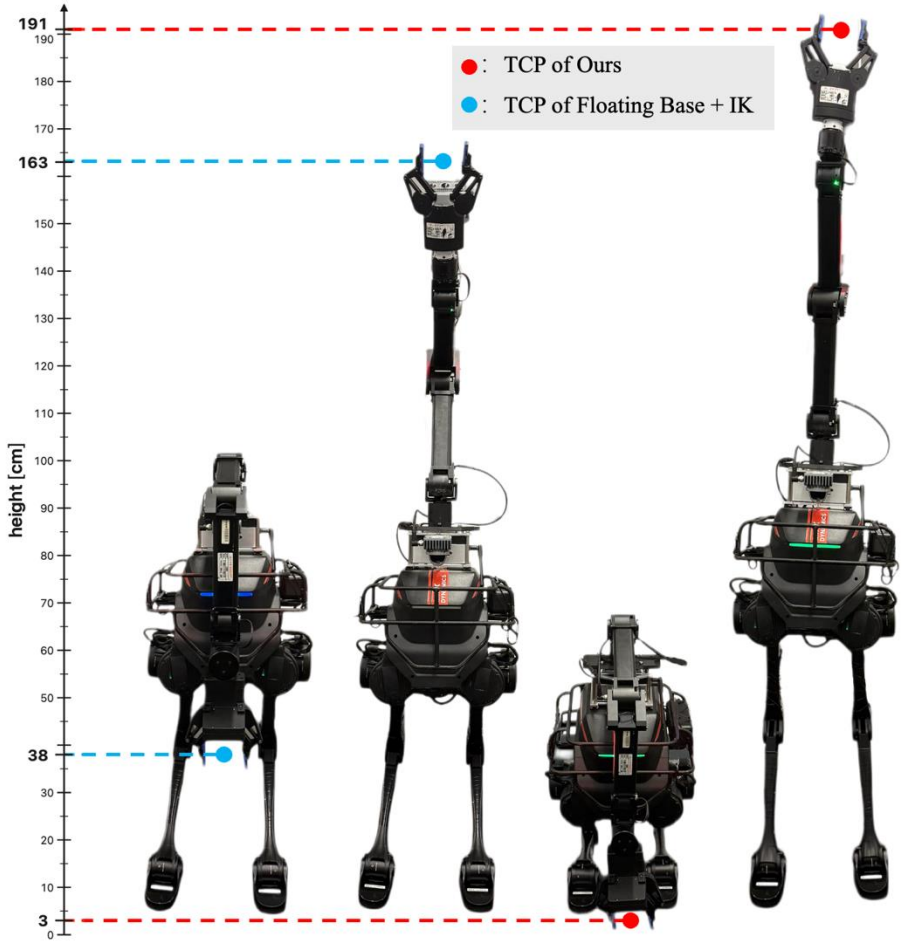
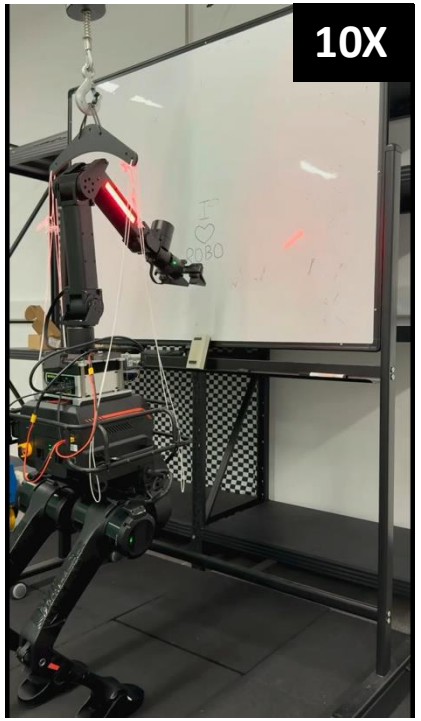
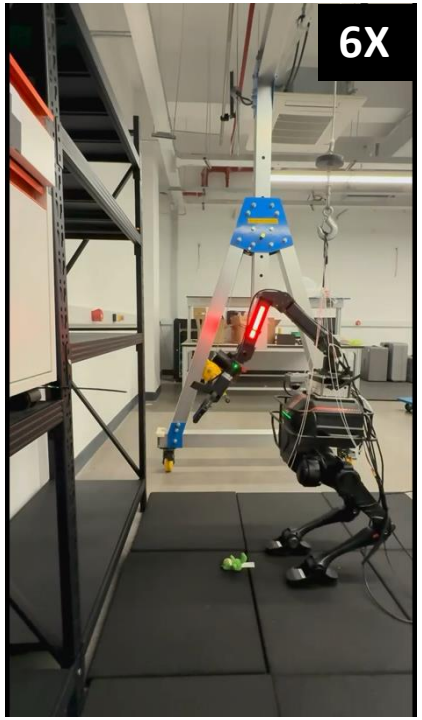


Fig2. Workspace



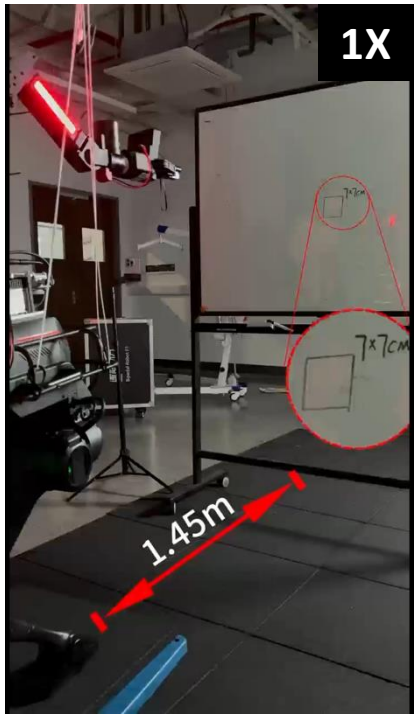
Board Wiping



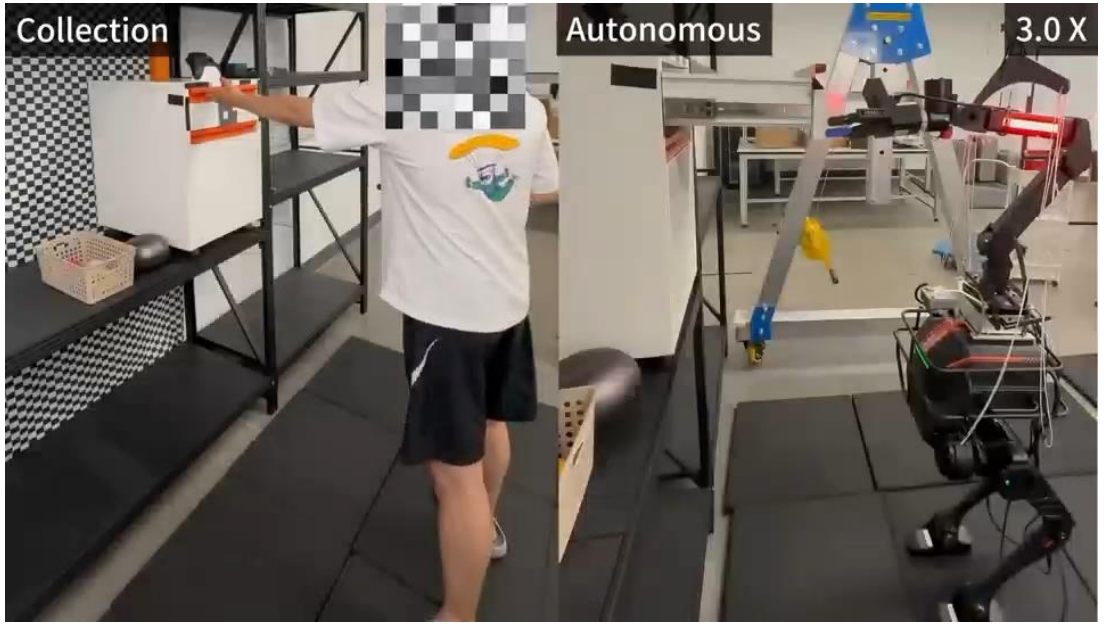
Vertical Placement



Pick-and-Place



Chicken-head Stabilization



Diffusion Policy Integration